Formal Verification of Lamport's Mutex algorithm for N processes.

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March 11, 2026

1 Introduction

The mutex problem, as originally posed by Dijkstra in [1], is a fundamental result in concurrent computing.

1.1 Problem Statement

blah blah

1.2 Formal verification in Lean4

blah blah

2 Methodology

This algorithm will be formally verified using a mathematical model representative of the system. The system will be modelled as a state machine, with each process moving between three states. These state transitions will be proven to uphold an invariant, ensuring that specific conditions are met. The Mutex property will be one of these conditions, that no two processes can be in the critical section of their computation at the same time.

3 Related literature

safdas

4 Risks

References

[1] E. W. Dijkstra. Solution of a problem in concurrent programming control. *Communications of the ACM*, 8(9):569, Sept. 1965.